A Comparative Study of Machine Learning and Deep Learning Models for Multi-Class Text Classification

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***Abstract*—Text classification is a fundamental task in natural language processing that involves assigning predefined categories to textual data. In this project, we conduct an extensive comparative analysis of traditional machine learning and deep learning models for a multi-class text classification problem involving ten distinct categories. The study evaluates the effectiveness of different word representation techniques, including TF-IDF, trainable embeddings, Skip-gram (Word2Vec), and pre-trained GloVe embeddings. Logistic Regression and a Deep Neural Network are trained using TF-IDF features, while multiple recurrent neural network architectures—including SimpleRNN, GRU, LSTM, and their bidirectional variants—are implemented using embedding-based representations. Experimental results demonstrate that Logistic Regression with TF-IDF achieves the best overall performance with a Macro F1-score of 0.6479, highlighting the continued effectiveness of traditional machine learning approaches for large-scale text classification tasks.**

***Index Terms*—Text Classification, TF-IDF, Logistic Regression, Deep Learning, RNN, GRU, LSTM, Word Embeddings**

I. Introduction

Text classification plays a critical role in many real-world applications such as question categorization, document indexing, sentiment analysis, and information retrieval. In a multi-class setting, the task becomes particularly challenging due to overlapping vocabulary and semantic similarity between classes.

Recent advances in deep learning have introduced powerful neural architectures capable of modeling sequential and contextual information. However, traditional machine learning models using sparse representations such as TF-IDF often remain competitive, especially for large and well-structured datasets.

The primary objective of this project is to empirically compare traditional machine learning and neural network-based approaches under a unified experimental framework.

II. Related Work and Background

A. Sparse Word Representations

TF-IDF emphasizes discriminative words but does not capture semantic similarity. Despite this limitation, it remains effective when strong lexical signals exist.

B. Dense Word Embeddings

Word2Vec and GloVe map words into dense vector spaces, capturing semantic and syntactic relationships.

C. Sequential Neural Models

RNNs process text sequentially but suffer from vanishing gradients. LSTM and GRU architectures address this issue using gating mechanisms.

III. Methodology

A. Dataset Description

The dataset consists of question-answer text samples categorized into ten classes. Training and testing sets contain 93,333 and 59,999 samples respectively, with no missing or duplicate entries.

B. Exploratory Data Analysis

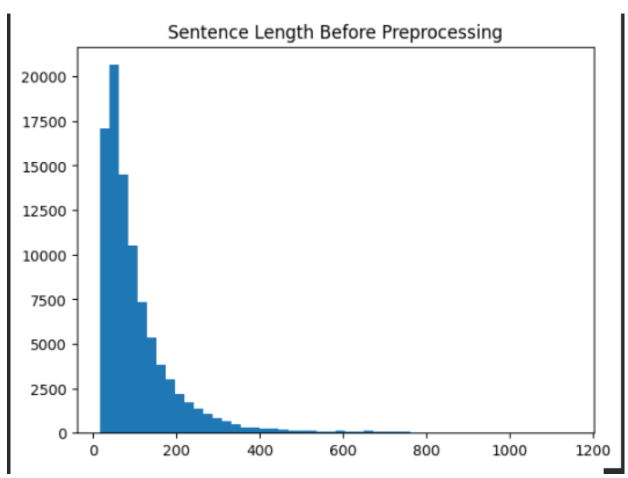

Fig. 1. Distribution of sentence lengths before preprocessing

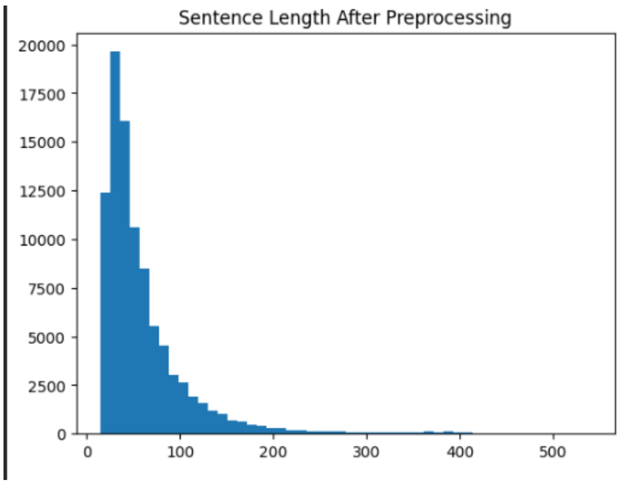

Fig. 2. Distribution of sentence lengths after preprocessing

Sentence length analysis guided the selection of padding length for neural models.

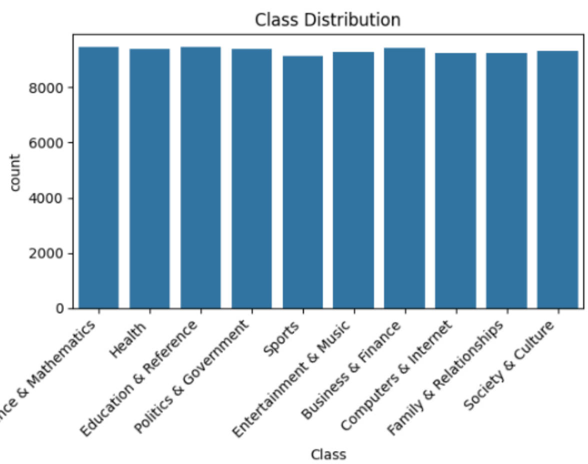

Fig. 3. Class distribution of the training dataset

The dataset is highly balanced, motivating the use of Macro F1-score.

C. Text Preprocessing Pipeline

Preprocessing steps include:

- Lowercasing text
- Removing punctuation and non-alphabetic characters
- Removing English stopwords

D. Word Representation Techniques

1) TF-IDF: TF-IDF was used to generate sparse vectors for Logistic Regression and the Deep Neural Network.

2) Trainable Embeddings: Neural models used trainable embedding layers learned during training.

3) Skip-gram (Word2Vec): Skip-gram embeddings were trained on the corpus to capture contextual relationships.

4) GloVe Embeddings: Pre-trained GloVe embeddings were used to incorporate external semantic knowledge.

E. Model Architectures

1) Logistic Regression with TF-IDF: Logistic Regression was used as the primary baseline model. Trained on TF-IDF features using a one-vs-rest strategy, it efficiently learned linear decision boundaries in high-dimensional space. Due to strong class-specific keywords, this model achieved the highest Macro F1-score among all evaluated models.

2) Deep Neural Network with TF-IDF: A fully connected Deep Neural Network was trained using TF-IDF features to introduce non-linearity. The architecture consists of dense layers with ReLU activation followed by a softmax output layer. Despite increased complexity, the DNN did not outperform Logistic Regression.

3) Recurrent Neural Network Models: SimpleRNN, GRU, and LSTM models were implemented using embedding-based inputs. Bidirectional variants were also explored to capture contextual information from both directions.

F. Training Strategy

Neural models were trained using the Adam optimizer and Sparse Categorical Crossentropy loss. Manual hyperparameter tuning was performed by varying hidden units (64 and 128). Dropout and Early Stopping were applied to reduce overfitting.

IV. Results and Analysis

A. Overall Model Performance

TABLE I
PERFORMANCE COMPARISON OF MODELS

Model	Accuracy	Macro F1
Logistic Regression (TF-IDF)	**0.6495**	**0.6479**
Deep Neural Network (TF-IDF)	0.5922	0.5941
SimpleRNN	0.5011	0.4820
GRU	0.5525	0.5425
LSTM	0.5603	0.5549
Bidirectional SimpleRNN	0.2746	0.2513
Bidirectional GRU	0.5544	0.5511
Bidirectional LSTM	0.5457	0.5516

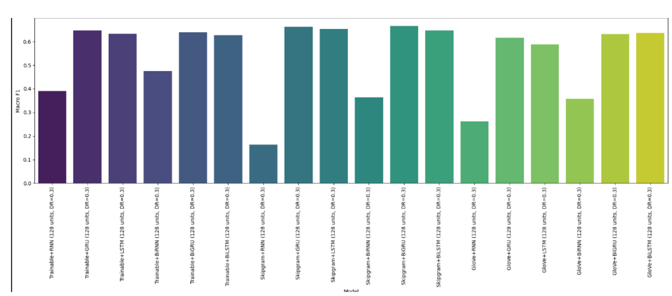

Fig. 4. Comparison of model performance using Macro F1-score

B. Discussion of Results

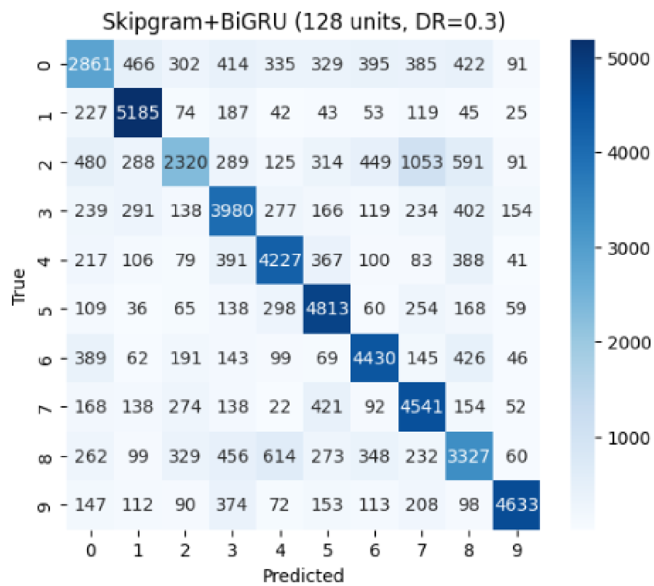

Fig. 5. Confusion matrix of Logistic Regression with TF-IDF

Logistic Regression with TF-IDF achieved the best performance overall. Neural models, although more expressive, struggled to outperform the linear baseline.

The Bidirectional SimpleRNN performed poorly due to vanishing gradient issues, while GRU and LSTM models demonstrated improved stability.

C. Why TF-IDF Outperformed Neural Models

TF-IDF directly highlights discriminative keywords. Neural models must jointly learn embeddings and classifiers, which can dilute learning signals when lexical cues alone are sufficient.

V. Limitations

Neural models were trained with limited hyperparameter ranges and epochs. Attention mechanisms and contextualized embeddings were not explored.

VI. Future Work

Future work may explore transformer-based architectures such as BERT and ensemble approaches combining TF-IDF and neural representations.

VII. Conclusion

This project presented a comprehensive comparison of machine learning and deep learning models for multi-class text classification. Experimental results demonstrate that Logistic Regression with TF-IDF remains highly effective for large-scale, balanced datasets.

References

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